

Smart grid and renewable energy:

Grafana is a useful tool for renewable energy systems and smart grid monitoring and optimisation. Energy providers can effectively manage the production, distribution, and consumption of electricity because of its real-time display, historical analysis, and alerting features.

Comparison of Grafana with other IoT visualisation tools

Grafana is a powerful visualisation tool for IoT based applications. Several other tools are also available that provide similar features and functionalities. However, all these differ in various ways. Table 1 compares Grafana with four other tools based on a few salient parameters.

Limitations of Grafana

Grafana can be difficult for novices to learn because it requires familiarity with data sources, queries, and a visualisation setup. Advanced customisation requires knowledge of scripting and

JSON technologies. Some advanced visualisation may not work for all categories of data.

Grafana does not store the data, and depends on external databases to do so. Numerous intricate panels or data points on a dashboard can cause it to lag and use more resources. Community plugins might not be regularly maintained and frequently lack thorough documentation.

It has some scaling challenges as the number of users increases.

Grafana is a strong and flexible

visualisation tool that works especially well for observability and real-time monitoring. It works best when combined with contemporary data sources because it provides a great deal of scalability and customisation. Alternative tools like ThingsBoard (for IoT), Kibana (for Elasticsearch), or Node-RED (for workflow orchestration) may be more appropriate in some cases as Grafana has a higher learning curve. Freeboard may be adequate for dashboards that are lightweight. **END** 

References

- <https://grafana.com/oss/grafana/> [Overview of Grafana]
- <https://scaleyourapp.com/what-is-grafana-why-use-it-everything-you-should-know-about-it/> [Technical aspects of Grafana]
- <https://grafana.com/oss/grafana/> [Documentation of Grafana]
- <https://www.craigirisi.com/post/pros-and-cons-of-different-tools-grafana> [Pros and cons of Grafana]

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Domination Set: [8, 9]
Domination Number: 2

Applications and insights

Toeplitz networks offer a wide range of practical applications due to their organised structure and computational efficiency. In the realm of communication networks, they are instrumental in modelling signal processing systems where periodicity and symmetry play crucial roles, facilitating effective resource allocation and data transmission. In optimisation scenarios, Toeplitz graphs assist in minimising costs while ensuring comprehensive coverage of network nodes, applicable in contexts such as sensor placement and power grid management. Furthermore, their translational invariance proves beneficial for addressing challenges in image processing, DNA sequencing, and time-series analysis. Utilising SageMath, researchers and professionals can effectively model these networks, compute dominating sets, and derive significant insights into their structural

characteristics, paving the way for advancements in network design and analysis.

Toeplitz networks help converge mathematical theory and practical applications, delivering sophisticated solutions for challenges in areas such as communication, optimisation, and data analysis. Investigating domination within these networks yields valuable insights into their architecture and functionality, especially in terms of resource optimisation and improving network performance. SageMath stands out as an effective instrument, allowing researchers to effortlessly model, analyse, and visualise Toeplitz networks. By connecting theoretical concepts with practical implementation, SageMath enables users to fully harness the capabilities of graph theory, promoting innovation and enhancing our comprehension of structured networks. **END** 

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